

From Data Insight to Prediction

Predicting NYC Airbnb nightly prices with statistical testing and Ridge regression

DASC 4850

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Dataset	New York City Airbnb Open Data (2019)
Original shape	48,895 rows x 16 columns
Final shape	48,396 rows x 20 columns
Target variable	price (model fit on log_price, reported back in USD)
Statistical question	Do entire-home listings cost more than private rooms?

Project overview

This project uses the NYC Airbnb listings dataset to explain what drives nightly price and to build a predictive model. The analysis combines data cleaning, exploratory visualization, one statistical test, and a regularized regression model.

Data wrangling and feature engineering

- Removed duplicate rows and filtered invalid records with non-positive price.
- Restricted minimum_nights to the range 1 to 365 to remove unrealistic listings.
- Removed extreme price outliers above the 99th percentile (prices greater than \$799).
- Filled missing reviews_per_month values with 0.
- Converted last_review to datetime and created days_since_last_review.
- Created has_reviews and host_is_multi_listing.

Exploratory analysis

The figures below show a strongly right-skewed price distribution, a large room-type effect, and much higher average prices in Manhattan.

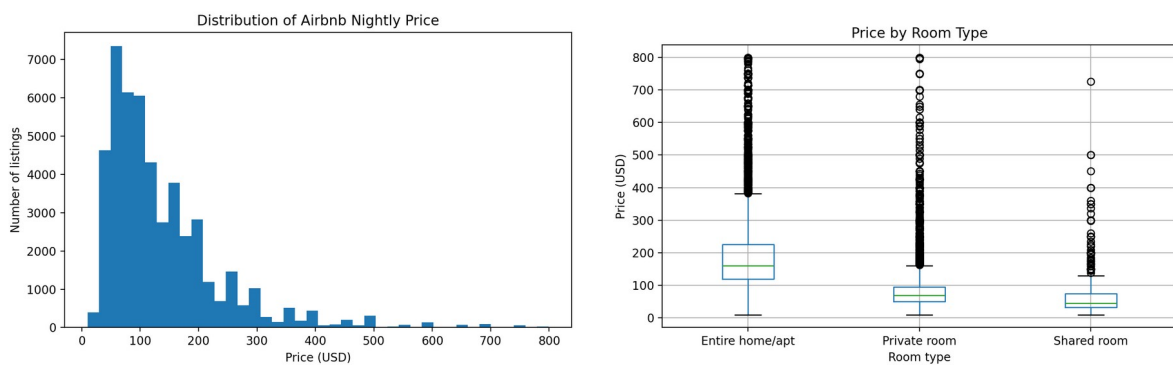


Figure 1. Price distribution is heavily right-skewed. Figure 2. Entire home/apartment listings have a much higher median price than private rooms and shared rooms.

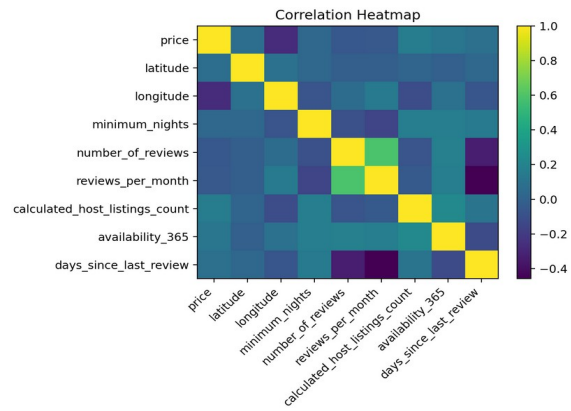
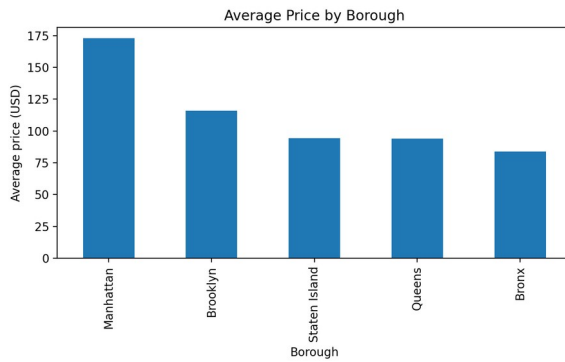


Figure 3. Manhattan has the highest average price. Figure 4. Numeric correlations with price are modest, so categorical features such as room type and borough add important information.

Selected descriptive statistics

Stat	Price	Min nights	Reviews	Rev/month	Avail.365
Mean	137.58	6.79	23.42	1.1	112.04
Std	103.78	16.13	44.71	1.6	131.23
Median	105.0	3.0	5.0	0.38	44.0

Statistical question and test

I compared the mean price of entire-home listings and private rooms using Welch's two-sample t-test. This test is appropriate because the groups are independent and do not need equal variances.

Group	n	Mean price (USD)
Entire home/apt	25,013	189.10
Private room	22,233	83.40

The mean difference was \$105.70 (95% CI \$104.14 to \$107.27), with $p < 0.001$ and Cohen's $d = 1.18$. Entire-home listings are substantially more expensive than private rooms.

Predictive modelling

I built a Ridge regression model to predict nightly price. Numeric variables were scaled, categorical variables were one-hot encoded, and the data was split into 60% training, 20% validation, and 20% test sets. The model was trained on `log_price` to reduce skewness, but predictions were converted back to dollars for evaluation.

Validation tuning results

alpha	val RMSE	val MAE
0.01	84.205	48.892
0.1	84.205	48.892
1.0	84.207	48.892
10.0	84.23	48.895
50.0	84.328	48.917
100.0	84.438	48.954

Final model performance

Split	RMSE	MAE
Validation	84.20	48.89
Test	82.83	47.58

Interpretation and insight

The strongest coefficients support the exploratory analysis. Entire-home listings have a positive coefficient, shared rooms have a negative coefficient, and Manhattan is positively associated with price. These results suggest that accommodation type and location are major determinants of Airbnb pricing in NYC.

Feature	Coefficient
neighbourhood_group_Staten Island	-0.660
room_type_Entire home/apt	0.622
room_type_Shared room	-0.507
neighbourhood_group_Manhattan	0.353
neighbourhood_group_Queens	0.171

Limitation and recommendation

A key limitation is that the model does not include amenity detail, listing descriptions, host response behavior, or image quality. Those missing features likely explain part of the remaining prediction error. A practical next step is to compare this baseline against Random Forest or XGBoost.

AI Use Appendix

AI tools were used for project planning, notebook structure, and wording support. All numbers in the report were verified by running the code, checking the saved tables, and confirming that the final notebook executed from top to bottom.